

Bayesian Cognition

Probabilistic models of action, perception,
inference, decision and learning

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Schedule

- C1 : mardi 18/11, 13h30-15h30 campus C-TP05
- C2 : lundi 24/11, 15h45-17h45 campus C113
- C3 : mardi 25/11, 15h45-17h45 campus C113
- C4 : mercredi 26/11, 15h45-17h45 minatec M255
- C5 : mercredi 03/12, 15h45-17h45 minatec M254
- C6 : lundi 08/12, 15h45-17h45 campus C113
- C7 : jeudi 11/12, 10h30-12h30 campus C015
- C8 : lundi 15/12, 15h45-17h45 campus C113
- C9 : mercredi 17/12, 13h30-15h30 campus C015
- Exam
 - ??
 - une feuille A4 de notes recto-verso autorisée

Contents

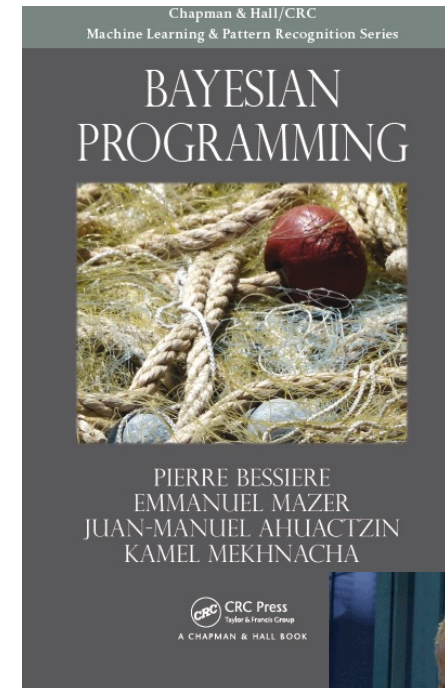
- Part I: Theoretical foundations and the Bayesian programming methodology
 - Classes 1, 2 and 3
- Part II: Bayesian Robot Programming
 - Classes 4 and 5
- Part III: Bayesian Cognitive Modeling
 - Classes 6, 7, half of 8
- Part IV: Bayesian Model Selection
 - End of class 8, class 9

Contents

- What this course is not
 - No in-depth presentation
 - of inference algorithms
 - EM, variational inference, message passing, etc.
 - of sampling methods
 - MCMC, Gibbs sampling, Metropolis-Hastings, etc.
 - of very specific models
 - EKF, MRF, RBM, deep learning, conv nets, SVM, RBF, etc.
 - Instead, survey of all these and their relationship
 - Not a Bayesian statistics course
 - Not a practical probabilistic programming course
 - Not a course on(ly on) Bayes-optimal cognitive models

Some reference books

- Jaynes, Probability Theory: The Logic of Science, 2003
 - See also Aubrey Clayton's video series "Probability Theory Lectures" on his YouTube channel
- MacKay, Information Theory, Inference and Learning Algorithms, 2003
- Knill & Richards, Perception as Bayesian Inference, 2008
- Dienes, Understanding Psychology as a Science: An introduction to scientific and statistical inference, 2008
- Murphy, Machine Learning: A Probabilistic Perspective, 2012
- **Bessière et al., Bayesian Programming, 2013**
- Lee & Wagenmakers, Bayesian cognitive modeling, 2013
- Clayton, Bernoulli's Fallacy: Statistical Illogic and the Crisis of Modern Science, 2021
- Ma, W. J., Kording, K. P., and Goldreich, D. (2023). Bayesian models of perception and action: An introduction. MIT Press, Cambridge, MA.
- Griffiths, T. L., Chater, N., and Tenenbaum, J. B. (2024). Bayesian models of cognition: Reverse engineering the mind. The MIT Press, Cambridge, MA; London, England.



To get more info

- Email:
 - julien.diard@univ-grenoble-alpes.fr
- Website:
 - <http://diard.wordpress.com>
 - Lebeltel et al., *Autonomous Robots*, 2004
 - Colas et al., *Acta Biotheoretica*, 2010
 - Variant in French: Colas et al., *Intellectica*, 2016
 - Diard et al., 2025, chapitre “Modélisation des processus langagiers”
 - Annales d’examens
 - Vidéo du cours 2020-2021 !